

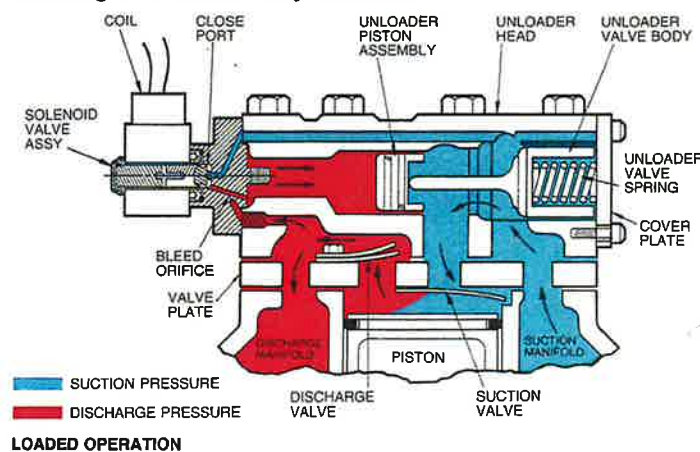
energized, cylinders load. Check solenoid valve for proper operation. An audible click can be heard when the valve actuates.

Capacity Control System – Electric

Electrically operated control valve is actuated by remote signal to the electric solenoid coil which has the same voltage as the unit control voltage. No manual adjustments to the electric unloader valve are necessary. When the solenoid is “de-energized” the passageway in the valve are aligned for “loaded” conditions. When the solenoid is “energized” the valve passageways are aligned for unloading.

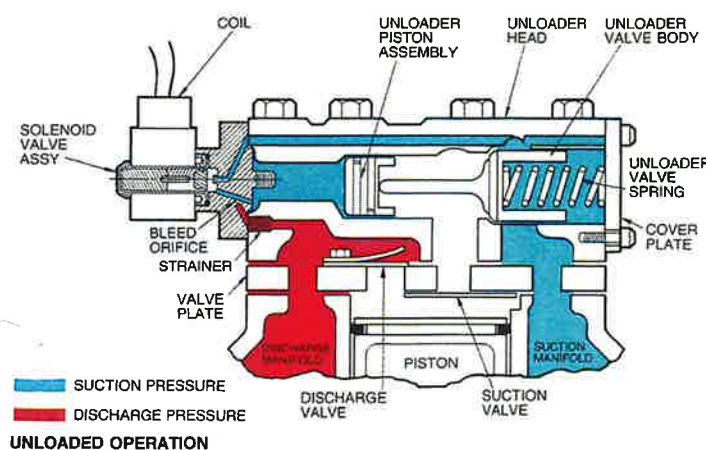
Loaded Operation

This capacity control valve is controlled by an electric solenoid. When the solenoid is de-energized the valve loads the cylinder bank (2 cylinders) as shown. When the suction pressure rises above a set point, an external controller de-energizes the solenoid coil. This closes the capacity control valve port, allowing discharge pressure to build-up behind the unloader piston assembly. A high enough pressure (25 to 40 psig above suction) will compress the unloader valve spring, opening the unloader suction port. Suction gas can now be drawn into the cylinder, running the bank fully loaded.



Unloaded Operation

As the suction pressure drops below the set point of the external controller, the solenoid coil is energized. This opens the capacity control valve port, allowing the discharge gas behind the unloader piston assembly to vent back to suction side. The unloader valve spring at this point can move the unloader valve body to the left, blocking the unloader suction port. The cylinder bank is now isolated from the compressor suction manifold, unloading these two cylinders. No refrigerant is allowed into cylinders and no compression takes place.

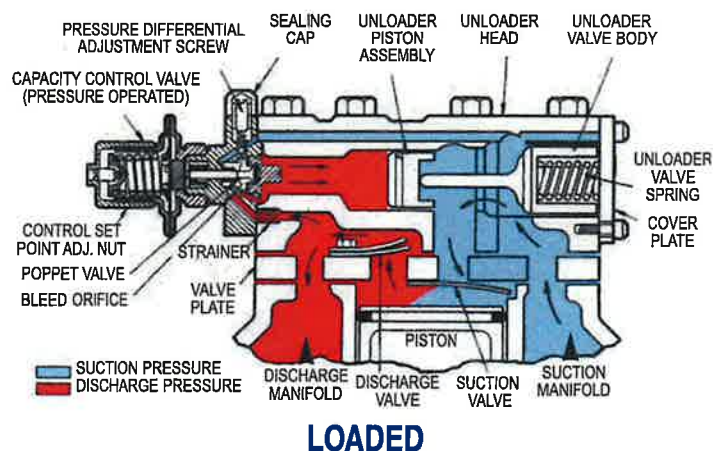


Capacity Control System – Pressure

Pressure-operated control valve is controlled by suction pressure and activated by discharge pressure. Each valve controls 2 cylinders (one bank). On start-up, controlled cylinders do not load up until differential between suction and discharge pressures is approximately 25 psig.

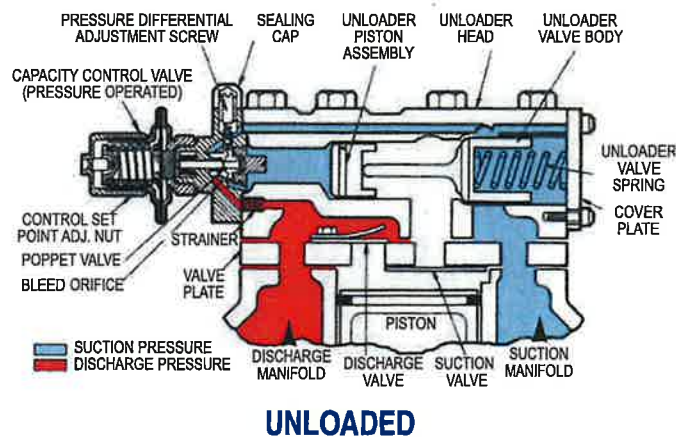
Loaded Operation

When suction pressure rises high enough to overcome control set point spring, the diaphragm snaps to the left and relieves pressure against the poppet valve. The drive spring moves the poppet valve to left and it seats in the closed position. With poppet valve closed, discharge gas is directed into the unloader-piston chamber and pressure builds up against the piston. When pressure against unloader valve spring, piston moves valve to the right, opening suction port. Suction gas can now be drawn into cylinders and the bank is running fully loaded.



Unloaded Operation

As suction pressure drops below set point, control spring expands, snapping diaphragm to right. This forces poppet valve open and allows gas from discharge manifold to vent through base of control valve to suction side. Loss of full discharge pressure against unloaded piston allows unloader valve spring to move valve left to closed position. The suction port is blocked, isolating the cylinder bank from the suction manifold. The cylinder bank is now unloaded.



Adjustments

Control Set Point (cylinder load-up point) is adjustable from 0 psig to 86 psig. Refer to fig. 1 & 2. Turn adjustment nut clockwise to the bottom stop. In this position, the cylinder load up pressure is 86 psig. Control set point is regulated to desired pressure by turning the adjustment nut counterclockwise. The number of turns can be determined from the curve. Each full turn counterclockwise decreases the load-up point by approximately 7.2 psi. Approximately 12 turns changes the pressure from 85 psi to 0 psi.

Pressure Differential between cylinder load-up point and unload point is adjustable from 6 psi to 16psi. Refer to fig. 1 & 3. Turn adjustment screw counterclockwise to the backstop. In this position, the differential is 6 psi. Differential is set by turning the adjustment screw clockwise. The number of turns to the desired differential can be determined from the curve. Each full turn clockwise increases the differential by approximately 0.8 psi. Approximately 5 turn changes the differential 6 psi to 10 psi.

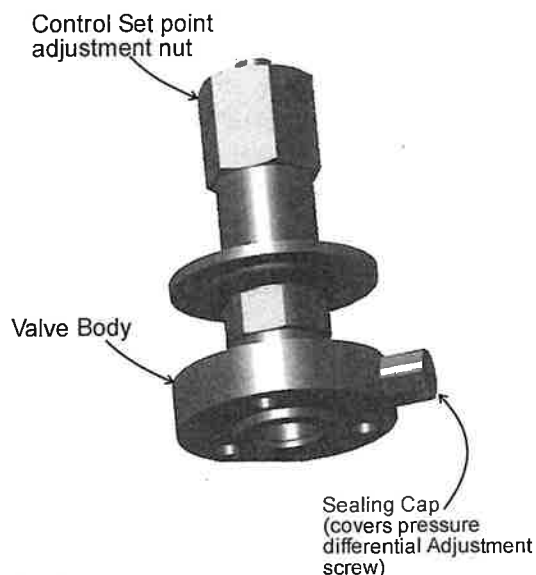


Fig. #1 Pressure-Operated Capacity Control Valve

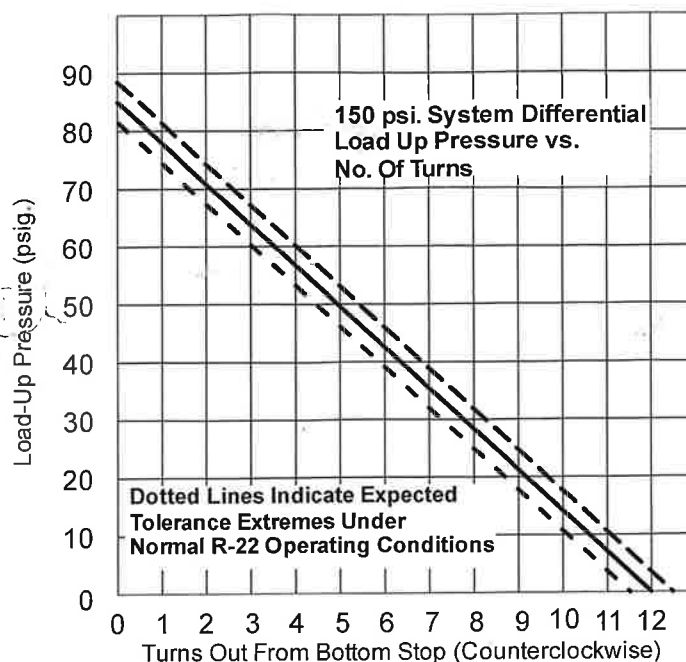


Fig. #2 Control Set Point

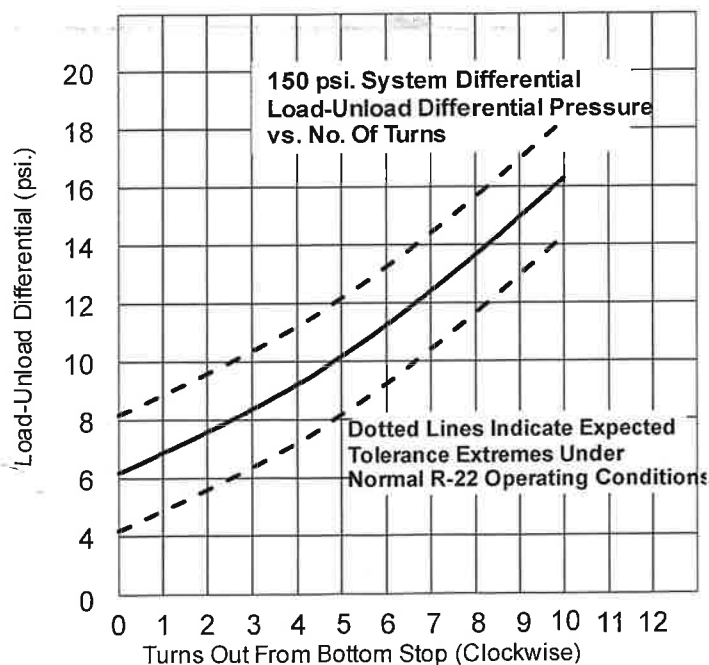


Fig. #3 Differential Set Point

06D Notes:

Once the old style cylinder head has been removed, the following procedures should be followed to insure proper installation and operation of the unloader head:

- 1) Make sure all of the old gasket material is removed.
- 2) Install the new cylinder head gasket on the valve plate assembly
- 3) Install the cylinder head assembly. Torque the cylinder head bolts to 30 – 35 ft. lbs.
- 4) New valve plate gasket is supplied in the event the original valve plate gasket is damaged or if the valve plate assembly is replaced, and it will fit the majority of 06D compressors.
- 5) The valve plate gasket included in the kit will not fit the old 13 cfm compressors (06DR013 & 06DM313 built before 1985). Valve plate gasket 6D40-1073 must be ordered separately. It will fit newer models.

Note: To re-use a hot gas bypass unloading valve plate for suction cut-off unloading, remove the check valve assembly on the bottom of the valve plate by removing the 3 slotted screws. A new valve plate gasket will be required.

06E Notes:

- 1) Cylinder head assemblies include the capacity control valve. For the electrically actuated head assembly, the customer must purchase the coil separately. For additional voltage selections, see note 5.
- 2) Valve plate gaskets are supplied in the event the original valve plate gasket is damaged when removing the cylinder head. One valve plate gasket has a thickness of 0.32" to be used on 06ER/EY refrigeration models only. The other gasket has a thickness of 0.07" can be used on all other models.
- 3) Once the existing cylinder head has been removed, the stated procedure should be followed to insure proper installation and operation of the unloader head.
 - a) Remove standard cylinder from compressor. From 4 cylinder 06E model remove the right hand head, as viewed from the pump end. On 6 cylinder models, either side bank can accommodate an unloader cylinder head.
 - b) Make sure all old gasket material is removed
 - c) Install new cylinder head gasket on valve plate assembly using the cap screw and washer provided torque to 4 to 6 ft. lbs. If valve plate is moved during unloader head assembly process, ensure suction valves under valve plate remain properly positioned on dowel pins.
 - d) Install new unloader head assembly using existing bolts and tighten evenly by increments with final torque to be 90 – 100 ft. lbs.
 - e) Pressure actuated control valve must be adjusted, see adjustment instructions.
- 4) On older compressors with hot gas unloaders the old hot gas bypass unloading valve can be reused by removing the check valve and housing from the valve plate itself and reinstall using new applicable valve plate gasket